Classical theories of information science

信息论选讲之一

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Outline

1. Definition and Compacity
2. The Channel Coding Theorem
3. The Converse Proof
4. The Direct Proof

2. The Channel Coding Theorem

The Channel Coding Theorem

- We will justify the definition of the capacity of a Discrete Memoryless Channel (DMC) by the proving *the channel coding theorem*. This theorem, which consists of two parts

- **Direct Part**
  - Information can be communicated through a DMC with an arbitrarily small probability of error at any rate less than the channel capacity.

- **Converse**
  - If information is communicated through a DMC at a rate higher than the capacity, then the probability of error is bounded away from zero.

Definition of a Channel Code

Definition 7.9 An (n, M) code for a discrete memoryless channel with input alphabet $\mathcal{X}$ and output alphabet $\mathcal{Y}$ is defined by an encoding function

$$f : \{1, 2, \cdots, M\} \rightarrow \mathcal{X}^n$$

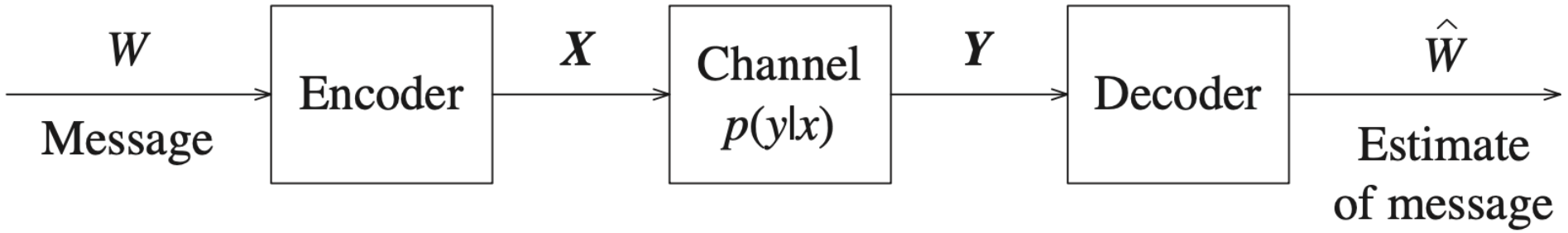
and a decoding function

$$g : \mathcal{Y}^n \rightarrow \{1, 2, \cdots, M\}.$$

- **Message Set** $\mathcal{W} = \{1, 2, \cdots, M\}$
- **Codewords** $f(1), f(2), \cdots, f(M)$
- **Codebook** The set of all codewords.

Assumptions and Notations

- W is randomly chosen from the message set $\mathcal{W}$, so $H(W) = \log M$.
- $\mathbf{X} = (X_1, X_2, \cdots, X_n)$; $\mathbf{Y} = (Y_1, Y_2, \cdots, Y_n)$
- Thus $\mathbf{X} = f(W)$.
- Let $\hat{W} = g(\mathbf{Y})$ be the estimate on the message W by the decoder.



Error Probabilities

Definition 7.10 For all $1 \leq w \leq M$, let

$$\lambda_w = \Pr\{\hat{W} \neq w | W = w\} = \sum_{\mathbf{y} \in \mathcal{Y}^n : g(\mathbf{y}) \neq w} \Pr\{\mathbf{Y} = \mathbf{y} | \mathbf{X} = f(w)\}$$

be the conditional probability of error given that the message is w.

Definition 7.11 The maximal probability of error of an (n, M) code is defined as

$$\lambda_{max} = \max_w \lambda_w.$$

Definition 7.12 The average probability of error of an (n, M) code is defined as

$$P_e = \Pr\{\hat{W} \neq W\}.$$

P_e vs λ_{max}

- The average probability of error P_e
- The conditional probability of error λ_w

$$
\begin{aligned}
P_e &= \Pr\{\hat{W} \neq W\} \\
&= \sum_w \Pr\{W = w\} \Pr\{\hat{W} \neq W | W = w\} \\
&= \sum_w \frac{1}{M} \Pr\{\hat{W} \neq w | W = w\} \\
&= \frac{1}{M} \sum_w \lambda_w,
\end{aligned}
$$

i.e., P_e is the arithmetic mean of λ_w, $1 \leq w \leq M$. It then follows that

$$P_e \leq \lambda_{max}.$$

Rate of a Channel Code

Definition 7.13 The rate of an (n, M) channel code is $n^{-1} \log M$ in bits per use.

Definition 7.14 A rate R is (asymptotically) achievable for a discrete memoryless channel if for any $\epsilon > 0$, there exists for sufficiently large n an (n, M) code such that

$$\frac{1}{n} \log M > R - \epsilon$$

and

$$\lambda_{max} < \epsilon.$$

The Channel Coding Theorem

Theorem 7.15 (Channel Coding Theorem) A rate R is achievable for a discrete memoryless channel if and only if $R \leq C$, the capacity of the channel.

- **Direct Part**
  - Information can be communicated through a DMC with an arbitrarily small probability of error at any rate less than the channel capacity.

- **Converse (逆命题)**
  - If information is communicated through a DMC at a rate higher than the capacity, then the probability of error is bounded away from zero.

3. The Converse Proof

The Converse

- The communication system consists of the r.v.'s

$$W, X_1, Y_1, X_2, Y_2, \cdots, X_n, Y_n, \hat{W}$$

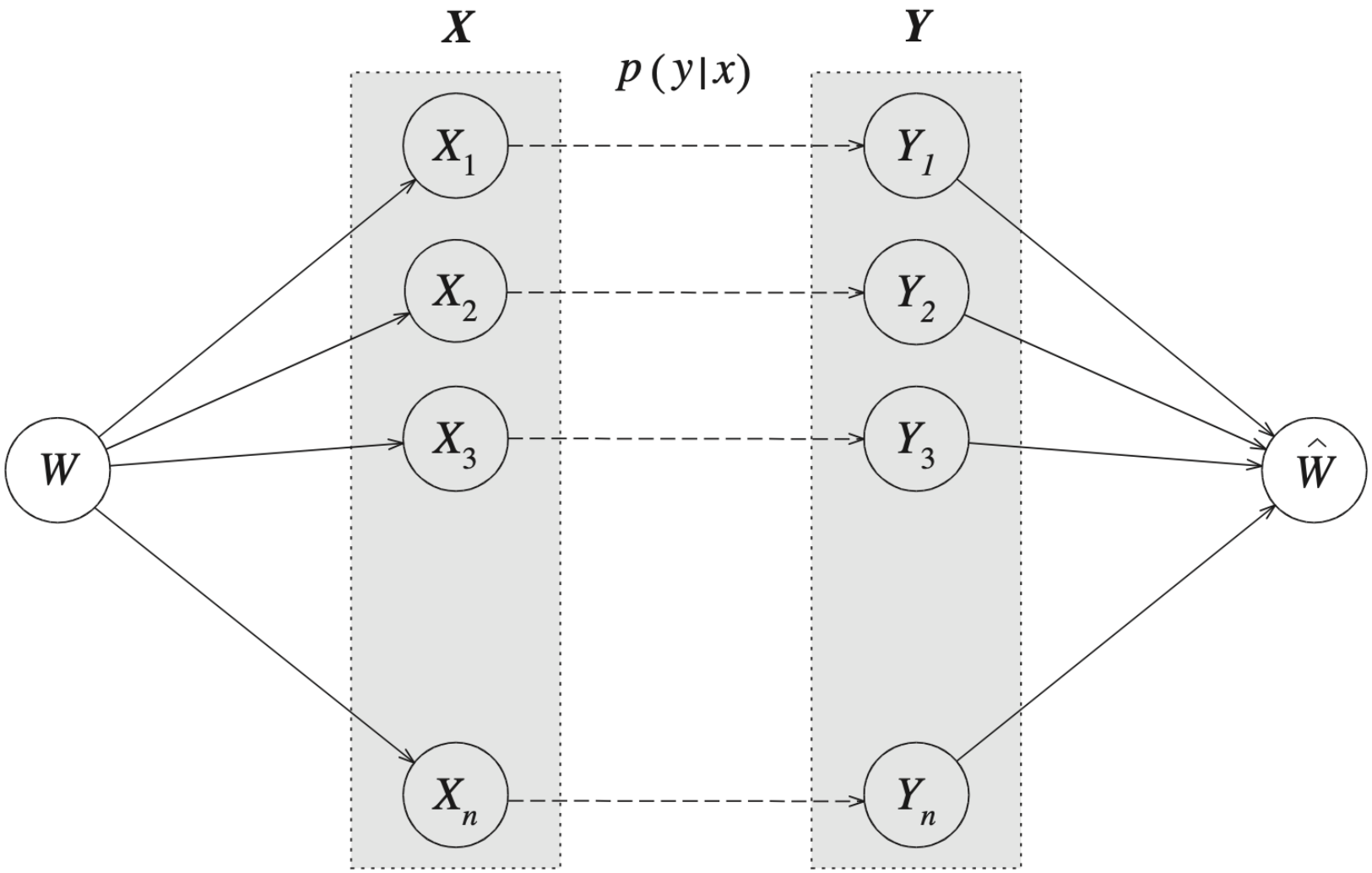
  generated in this order.

- The memorylessness of the DMC imposes the following Markov constraint for each i:

$$(W, X_1, Y_1, \cdots, X_{i-1}, Y_{i-1}) \to X_i \to Y_i$$

- The dependency graph can be composed accordingly.

The dependency graph



Markov chain

Proposition 2.9 $X_1 \to X_2 \to \cdots \to X_n$ forms a Markov chain if and only if

$$p(x_1, x_2, \cdots, x_n) = f_1(x_1, x_2) f_2(x_2, x_3) \cdots f_{n-1}(x_{n-1}, x_n)$$

for all $x_1, x_2, \cdots, x_n$ such that $p(x_2), p(x_3), \cdots, p(x_{n-1}) > 0$.

- Use q to denote the joint distribution and marginal distributions of all r.v.'s.
- For all $(w, \mathbf{x}, \mathbf{y}, \hat{w}) \in \mathcal{W} \times \mathcal{X}^n \times \mathcal{Y}^n \times \hat{\mathcal{W}}$ such that $q(\mathbf{x}) > 0$ and $q(\mathbf{y}) > 0$,

$$q(w, \mathbf{x}, \mathbf{y}\, \hat{w}) = q(w) \left(\prod_{i=1}^{n} q(x_i|w) \right) \left(\prod_{i=1}^{n} p(y_i|x_i) \right) q(\hat{w}|\mathbf{y}).$$

- $q(w) > 0$ for all w so that $q(x_i|w)$ are well-defined.
- $q(x_i|w)$ and $q(\hat{w}|\mathbf{y})$ are deterministic.
- The dependency graph suggests the Markov chain $W \to \mathbf{X} \to \mathbf{Y} \to \hat{W}$.
- This can be formally justified by invoking Proposition 2.9.

Compute the q(x,y)

Show that for $\mathbf{x}$ such that $q(\mathbf{x}) > 0$,

$$q(\mathbf{y}|\mathbf{x}) = \prod_{i=1}^{n} p(y_i|x_i)$$

First, for $\mathbf{x}$ and $\mathbf{y}$ such that $q(\mathbf{x}) > 0$ and $q(\mathbf{y}) > 0$,

$$\begin{aligned}
q(\mathbf{x}, \mathbf{y}) &= \sum_{w} \sum_{\hat{w}} q(w, \mathbf{x}, \mathbf{y}, \hat{w}) \\
&= \sum_{w} \sum_{\hat{w}} q(w) \left(\prod_{i} q(x_i|w) \right) \left(\prod_{i} p(y_i|x_i) \right) q(\hat{w}|\mathbf{y}) \\
&= \sum_{w} q(w) \left(\prod_{i} q(x_i|w) \right) \left(\prod_{i} p(y_i|x_i) \right) \sum_{\hat{w}} q(\hat{w}|\mathbf{y}) \\
&= \left[\sum_{w} q(w) \prod_{i} q(x_i|w) \right] \left[\prod_{i} p(y_i|x_i) \right]
\end{aligned}$$

Compute the q(x) and q(y|x) = q(x,y)/q(x)

Furthermore,

$$
\begin{aligned}
q(\mathbf{x}) &= \sum_{\mathbf{y}} q(\mathbf{x}, \mathbf{y}) \\
&= \sum_{\mathbf{y}} \left[\sum_{w} q(w) \prod_{i} q(x_i|w) \right] \left[\prod_{i} p(y_i|x_i) \right] \\
&= \left[\sum_{w} q(w) \prod_{i} q(x_i|w) \right] \left[\sum_{y_1} \sum_{y_2} \cdots \sum_{y_n} \prod_{i} p(y_i|x_i) \right] \\
&= \left[\sum_{w} q(w) \prod_{i} q(x_i|w) \right] \prod_{i} \left(\sum_{y_i} p(y_i|x_i) \right) \\
&= \sum_{w} q(w) \prod_{i} q(x_i|w)
\end{aligned}
$$

Therefore, for $\mathbf{x}$ such that $q(\mathbf{x}) > 0$,

$$
q(\mathbf{y}|\mathbf{x}) = \frac{q(\mathbf{x}, \mathbf{y})}{q(\mathbf{x})} = \prod_{i} p(y_i|x_i)
$$

Why C is related to I(X;Y)?

- $H(\mathbf{X}|W) = 0$ because $\mathbf{X}$ is a function of W.
- $H(\hat{W}|\mathbf{Y}) = 0$ because $\hat{W}$ is a function of $\mathbf{Y}$.
- Since W and $\hat{W}$ are essentially identical for reliable communication, assume

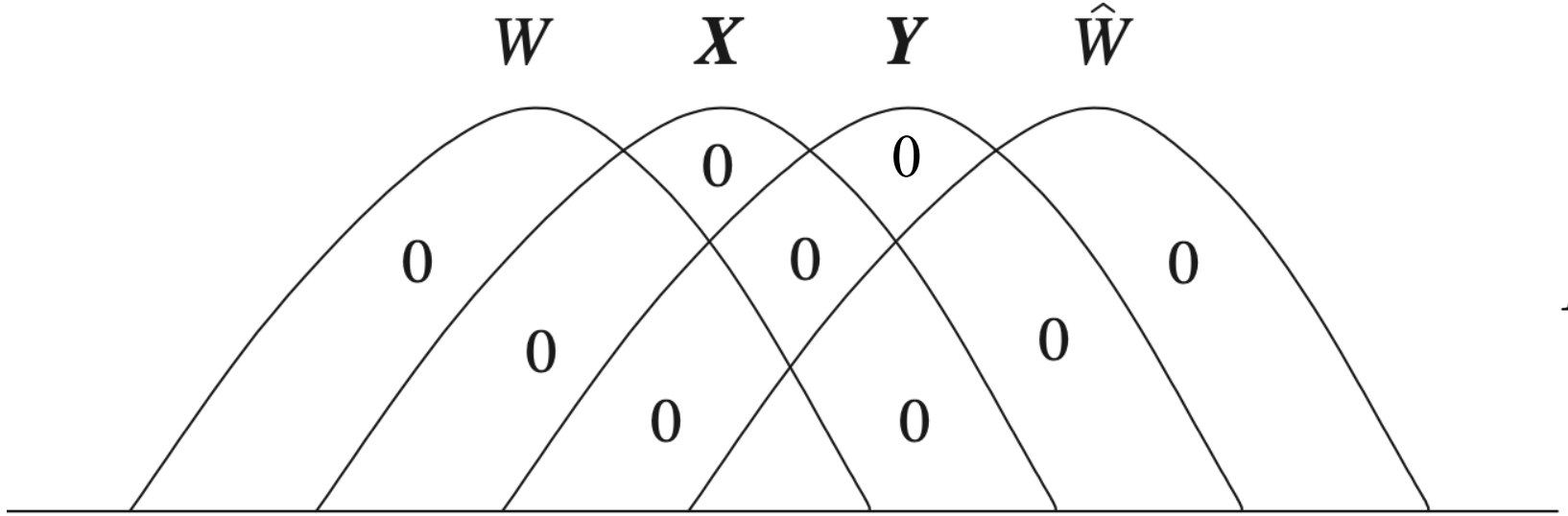
$$H(\hat{W}|W) = H(W|\hat{W}) = 0$$

- Then from the information diagram for $W \to \mathbf{X} \to \mathbf{Y} \to \hat{W}$, we see that

$$H(W) = I(\mathbf{X};\mathbf{Y}).$$

- This suggests that the channel capacity is obtained by maximizing $I(X;Y)$.

The information diagram for $W \to \mathbf{X} \to \mathbf{Y} \to \hat{W}$



Constraints

$$H(\mathbf{X}|W) = 0$$

$$H(\hat{W}|\mathbf{Y}) = 0$$

$$H(\hat{W}|W) = H(W|\hat{W}) = 0$$

- Then from the information diagram for $W \to \mathbf{X} \to \mathbf{Y} \to \hat{W}$, we see that

$$H(W) = I(\mathbf{X}; \mathbf{Y}).$$

Building blocks of the Converse

- For all $1 \leq i \leq n$,
$$I(X_i; Y_i) \leq C$$
- Then
$$\sum_{i=1}^{n} I(X_i; Y_i) \leq nC$$
- To be established in Lemma 7.16,
$$I(\mathbf{X}; \mathbf{Y}) \leq \sum_{i=1}^{n} I(X_i; Y_i)$$

- Therefore,
$$\begin{aligned} \frac{1}{n} \log M &= \frac{1}{n} H(W) \\ &= \frac{1}{n} I(\mathbf{X}; \mathbf{Y}) \\ &\leq \frac{1}{n} \sum_{i=1}^{n} I(X_i; Y_i) \\ &\leq C \end{aligned}$$

Proof of $I(\mathbf{X};\mathbf{Y}) \le \sum_{i=1}^{n} I(X_i;Y_i)$

Lemma 7.16. *For a discrete memoryless channel used with a channel code without feedback, for any $n \ge 1$,*

$$I(\mathbf{X};\mathbf{Y}) \le \sum_{i=1}^{n} I(X_i;Y_i), \tag{7.113}$$

where X_i and Y_i are, respectively, the input and the output of the channel at time i.

Proof. For any $(\mathbf{x},\mathbf{y}) \in \mathcal{X}^n \times \mathcal{Y}^n$, if $q(\mathbf{x},\mathbf{y}) > 0$, then $q(\mathbf{x}) > 0$ and (7.101) holds. Therefore,

$$q(\mathbf{y}|\mathbf{x}) = \prod_{i=1}^{n} p(y_i|x_i). \tag{7.101}$$

$$q(\mathbf{Y}|\mathbf{X}) = \prod_{i=1}^{n} p(Y_i|X_i) \tag{7.114}$$

holds for all $(\mathbf{x},\mathbf{y})$ in the support of $q(\mathbf{x},\mathbf{y})$. Then

$$-E \log q(\mathbf{Y}|\mathbf{X}) = -E \log \prod_{i=1}^{n} p(Y_i|X_i) = -\sum_{i=1}^{n} E \log p(Y_i|X_i), \tag{7.115}$$

or

$$H(\mathbf{Y}|\mathbf{X}) = \sum_{i=1}^{n} H(Y_i|X_i). \tag{7.116}$$

Hence,

$$\begin{aligned} I(\mathbf{X};\mathbf{Y}) &= H(\mathbf{Y}) - H(\mathbf{Y}|\mathbf{X}) && (7.117)\\ &\le \sum_{i=1}^{n} H(Y_i) - \sum_{i=1}^{n} H(Y_i|X_i) && (7.118)\\ &= \sum_{i=1}^{n} I(X_i;Y_i). && (7.119) \end{aligned}$$

The lemma is proved. □

Formal Converse Proof

1. Let R be an achievable rate, i.e., for any $\epsilon > 0$, there exists for sufficiently large n an (n, M) code such that

$$\frac{1}{n} \log M > R - \epsilon \quad \text{and} \quad \lambda_{max} < \epsilon$$

2. Consider

$$\begin{aligned}
\log M &\overset{a)}{=} H(W) \\
&= H(W|\hat{W}) + I(W;\hat{W}) \\
&\overset{b)}{\leq} H(W|\hat{W}) + I(\mathbf{X};\mathbf{Y}) \\
&\overset{c)}{\leq} H(W|\hat{W}) + \sum_{i=1}^{n} I(X_i;Y_i) \\
&\overset{d)}{\leq} H(W|\hat{W}) + nC,
\end{aligned}$$

We assume that a message W is randomly chosen from the message set $\mathcal{W}$ according to the uniform distribution. Therefore,

$$H(W) = \log M. \tag{7.80}$$

Theorem 2.42 (Data Processing Theorem). *If $U \to X \to Y \to V$ forms a Markov chain, then*

$$I(U;V) \leq I(X;Y). \tag{2.159}$$

Lemma 7.16. *For a discrete memoryless channel used with a channel code without feedback, for any $n \geq 1$,*

$$I(\mathbf{X};\mathbf{Y}) \leq \sum_{i=1}^{n} I(X_i;Y_i), \tag{7.113}$$

where X_i and Y_i are, respectively, the input and the output of the channel at time i.

$$\sum_{i=1}^{n} I(X_i;Y_i) \leq nC. \tag{7.107}$$

where

a) follows from (7.80);
b) follows from the data processing theorem since $W \to \mathbf{X} \to \mathbf{Y} \to \hat{W}$;
c) follows from Lemma 7.16;
d) follows from (7.107).

Formal Converse Proof

Theorem 2.47 (Fano's Inequality) Let X and $\hat{X}$ be random variables taking values in the same alphabet $\mathcal{X}$. Then

$$H(X|\hat{X}) \leq h_b(P_e) + P_e \log(|\mathcal{X}| - 1),$$

where h_b is the binary entropy function.

Corollary 2.48 $H(X|\hat{X}) < 1 + P_e \log |\mathcal{X}|$.

3. By Fano's inequality,

$$H(W|\hat{W}) < 1 + P_e \log M$$

4. Then,

$$\begin{aligned} \log M &< 1 + P_e \log M + nC \\ &\leq 1 + \lambda_{max} \log M + nC \\ &< 1 + \epsilon \log M + nC, \end{aligned}$$

Apply Fano's inequality to Step 2's result

$P_e \leq \lambda_{max}$. (average ≤ max)

$\lambda_{max} < \epsilon$. (Step 1: R achievable)

Therefore,

$$R - \epsilon < \frac{1}{n} \log M < \frac{\frac{1}{n} + C}{1 - \epsilon}$$

$\frac{1}{n} \log M > R - \epsilon$ (Step 1: R achievable)

5. Letting $n \to \infty$ and then $\epsilon \to 0$ to conclude that $R \leq C$.

Formal Converse Proof

$$
\begin{aligned}
\log M &\overset{a)}{=} H(W) \\
&= H(W|\hat{W}) + I(W;\hat{W}) \\
&\overset{b)}{\leq} H(W|\hat{W}) + I(\mathbf{X};\mathbf{Y}) \\
&\overset{c)}{\leq} H(W|\hat{W}) + \sum_{i=1}^{n} I(X_i;Y_i) \\
&\overset{d)}{\leq} H(W|\hat{W}) + nC,
\end{aligned}
$$

where

a) follows from (7.80);
b) follows from the data processing theorem since $W \to \mathbf{X} \to \mathbf{Y} \to \hat{W}$;
c) follows from Lemma 7.16;
d) follows from (7.107).

a)

We assume that a message W is randomly chosen from the message set $\mathcal{W}$ according to the uniform distribution. Therefore,

$$H(W) = \log M. \tag{7.80}$$

b)

Theorem 2.42 (Data Processing Theorem) If $U \to X \to Y \to V$ forms a Markov chain, then

$$I(U;V) \leq I(X;Y).$$

c)

Lemma 7.16. *For a discrete memoryless channel used with a channel code without feedback, for any $n \geq 1$,*

$$I(\mathbf{X};\mathbf{Y}) \leq \sum_{i=1}^{n} I(X_i;Y_i), \tag{7.113}$$

where X_i and Y_i are, respectively, the input and the output of the channel at time i.

d)

For a single transmission, we see from the definition of channel capacity that the mutual information between the input and the output cannot exceed the capacity of the channel, i.e., for all $1 \leq i \leq n$,

$$I(X_i;Y_i) \leq C. \tag{7.106}$$

Summing i from 1 to n, we have

$$\sum_{i=1}^{n} I(X_i;Y_i) \leq nC. \tag{7.107}$$

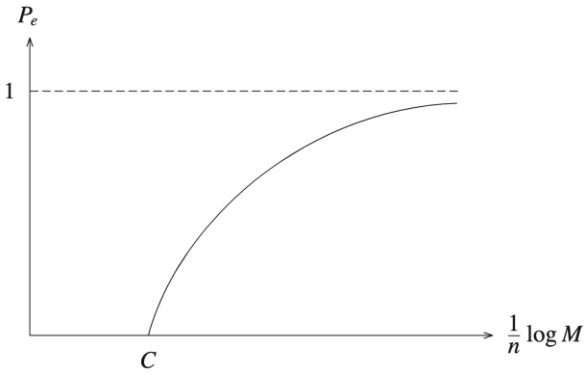
Asymptotic Bound for P_e: Weak Converse

Therefore, from (7.126),

$$\log M < 1 + P_e \log M + nC \tag{7.128}$$
$$\leq 1 + \lambda_{max} \log M + nC \tag{7.129}$$
$$< 1 + \epsilon \log M + nC, \tag{7.130}$$

- For large n,

$$P_e \geq 1 - \frac{1 + nC}{\log M} = 1 - \frac{\frac{1}{n} + C}{\frac{1}{n} \log M} \approx 1 - \frac{C}{\frac{1}{n} \log M}$$



- $\frac{1}{n} \log M$ is the actual rate of the channel code.
- If $\frac{1}{n} \log M > C$, then $P_e > 0$ for large n.
- This implies that if $\frac{1}{n} \log M > C$, then $P_e > 0$ for all n.

Strong Converse

- If there exists an $\epsilon > 0$ such that $\frac{1}{n} \log M \geq C + \epsilon$ for all n, then $P_e \to 1$ as $n \to \infty$.

4. The Direct Proof: Achievability

Achievability

- Consider a DMC $p(y|x)$.
- For every input distribution $p(x)$, prove that the rate $I(X;Y)$ is achievable by showing for large n the existence of a channel code such that
  1. the rate of the code is arbitrarily close to $I(X;Y)$;
  2. the maximal probability of error λ_{max} is arbitrarily small.
- Choose the input distribution $p(x)$ to be one that achieves the channel capacity, i.e., $I(X;Y) = C$.

Weak Asymptotic Equipartition Property (AEP)

We consider an information source $\{X_k, k \geq 1\}$ where X_k are i.i.d. with distribution $p(x)$. We use X to denote the generic random variable and $H(X)$ to denote the common entropy for all X_k, where $H(X) < \infty$. Let $\mathbf{X} = (X_1, X_2, \cdots, X_n)$. Since X_k are i.i.d.,

$$p(\mathbf{X}) = p(X_1)p(X_2)\cdots p(X_n). \tag{5.1}$$

Note that $p(\mathbf{X})$ is a random variable because it is a function of the random variables $X_1, X_2, \cdots, X_n$. We now prove an asymptotic property of $p(\mathbf{X})$ called the *weak asymptotic equipartition property* (weak AEP).

Theorem 5.1 (Weak AEP I).

$$-\frac{1}{n}\log p(\mathbf{X}) \to H(X) \tag{5.2}$$

in probability as $n \to \infty$, *i.e., for any* $\epsilon > 0$, *for* n *sufficiently large,*

$$\Pr\left\{\left|-\frac{1}{n}\log p(\mathbf{X}) - H(X)\right| \leq \epsilon\right\} > 1 - \epsilon. \tag{5.3}$$

Weak Asymptotic Equipartition Property (AEP)

Theorem 5.1 (Weak AEP I).

$$\mathbf{X} = (X_1, X_2, \cdots, X_n)$$

$$-\frac{1}{n}\log p(\mathbf{X}) \to H(X) \tag{5.2}$$

in probability as $n \to \infty$*, i.e., for any* $\epsilon > 0$*, for* n *sufficiently large,*

$$\Pr\left\{\left|-\frac{1}{n}\log p(\mathbf{X}) - H(X)\right| \le \epsilon\right\} > 1 - \epsilon. \tag{5.3}$$

Proof. Since $X_1, X_2, \cdots, X_n$ are i.i.d., by (5.1),

Since X_k are i.i.d.,

$$p(\mathbf{X}) = p(X_1)p(X_2)\cdots p(X_n).$$

$$-\frac{1}{n}\log p(\mathbf{X}) = -\frac{1}{n}\sum_{k=1}^{n}\log p(X_k). \tag{5.4}$$

The random variables $\log p(X_k)$ are also i.i.d. Then by the weak law of large numbers, the right hand side of (5.4) tends to

$$-E\log p(X) = H(X), \tag{5.5}$$

in probability, proving the theorem. □

Weak Asymptotic Equipartition Property (AEP)

Theorem 5.3 (Weak AEP II). *The following hold for any $\epsilon > 0$:*

1) If $\mathbf{x} \in W_{[X]\epsilon}^n$, then

$$2^{-n(H(X)+\epsilon)} \leq p(\mathbf{x}) \leq 2^{-n(H(X)-\epsilon)}. \tag{5.9}$$

2) For n sufficiently large,

$$\Pr\{\mathbf{X} \in W_{[X]\epsilon}^n\} > 1 - \epsilon. \tag{5.10}$$

3) For n sufficiently large,

$$(1-\epsilon)2^{n(H(X)-\epsilon)} \leq |W_{[X]\epsilon}^n| \leq 2^{n(H(X)+\epsilon)}. \tag{5.11}$$

The strongly typical set

We consider an information source $\{X_k, k \geq 1\}$ where X_k are i.i.d. with distribution $p(x)$. We use X to denote the generic random variable and $H(X)$ to denote the common entropy for all X_k, where $H(X) < \infty$. Let $\mathbf{X} = (X_1, X_2, \cdots, X_n)$.

Definition 6.1. *The strongly typical set $T^n_{[X]\delta}$ with respect to $p(x)$ is the set of sequences $\mathbf{x} = (x_1, x_2, \cdots, x_n) \in \mathcal{X}^n$ such that $N(x; \mathbf{x}) = 0$ for $x \notin \mathcal{S}_X$, and*

$$\sum_x \left| \frac{1}{n} N(x; \mathbf{x}) - p(x) \right| \leq \delta, \tag{6.1}$$

where $N(x; \mathbf{x})$ is the number of occurrences of x in the sequence $\mathbf{x}$, and δ is an arbitrarily small positive real number. The sequences in $T^n_{[X]\delta}$ are called strongly δ-typical sequences.

Strong AEP

Theorem 6.2 (Strong AEP). *There exists $\eta > 0$ such that $\eta \to 0$ as $\delta \to 0$, and the following hold:*

1) If $\mathbf{x} \in T^n_{[X]\delta}$, then

$$2^{-n(H(X)+\eta)} \le p(\mathbf{x}) \le 2^{-n(H(X)-\eta)}. \tag{6.2}$$

2) For n sufficiently large,

$$\Pr\{\mathbf{X} \in T^n_{[X]\delta}\} > 1 - \delta. \tag{6.3}$$

3) For n sufficiently large,

$$(1-\delta)2^{n(H(X)-\eta)} \le |T^n_{[X]\delta}| \le 2^{n(H(X)+\eta)}. \tag{6.4}$$

Lemma 7.17 Let $(\mathbf{X}', \mathbf{Y}')$ be n i.i.d. copies of a pair of generic random variables (X', Y'), where X' and Y' are independent and have the same marginal distributions as X and Y, respectively. Then

$$\Pr\{(\mathbf{X}', \mathbf{Y}') \in T^n_{[XY]\delta}\} \le 2^{-n(I(X;Y)-\tau)},$$

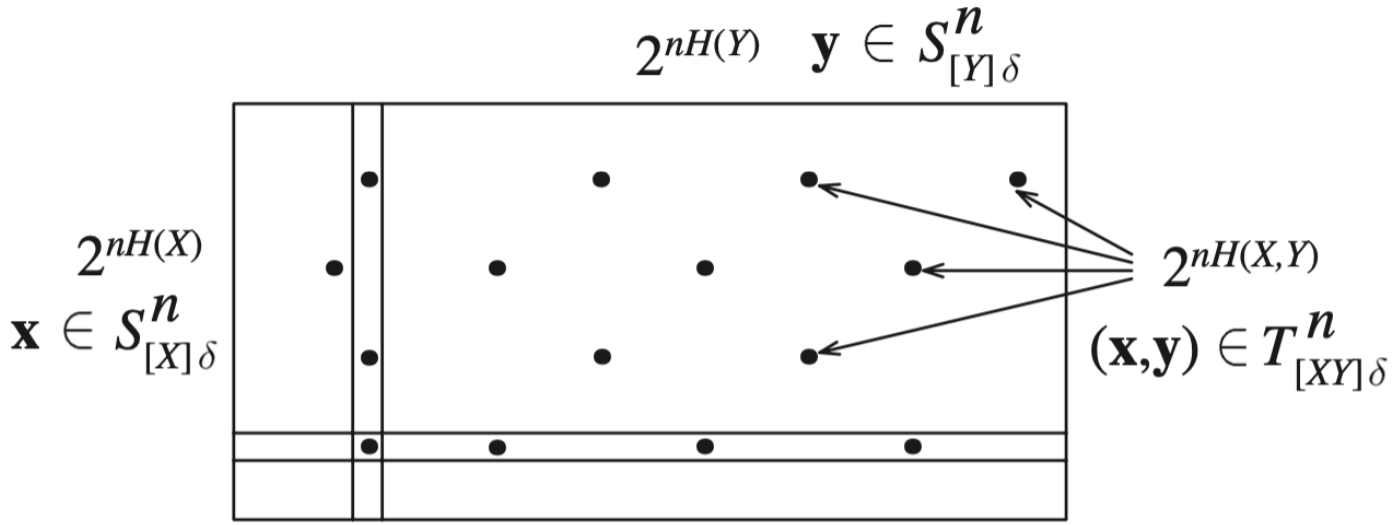
where $\tau \to 0$ as $\delta \to 0$.

Proof:

- Consider
$$\Pr\{(\mathbf{X}', \mathbf{Y}') \in T^n_{[XY]\delta}\} = \sum_{(\mathbf{x},\mathbf{y}) \in T^n_{[XY]\delta}} p(\mathbf{x})p(\mathbf{y})$$
- Consistency of strong typicality: $\mathbf{x} \in T^n_{[X]\delta}$ and $\mathbf{y} \in T^n_{[Y]\delta}$.
- Strong AEP: $p(\mathbf{x}) \le 2^{-n(H(X)-\eta)}$ and $p(\mathbf{y}) \le 2^{-n(H(Y)-\zeta)}$.
- Strong JAEP: $|T^n_{[XY]\delta}| \le 2^{n(H(X,Y)+\xi)}$.

- Then
$$\begin{aligned}
&\Pr\{(\mathbf{X}', \mathbf{Y}') \in T^n_{[XY]\delta}\} \\
&\quad \le 2^{n(H(X,Y)+\xi)} \cdot 2^{-n(H(X)-\eta)} \cdot 2^{-n(H(Y)-\zeta)} \\
&\quad = 2^{-n(H(X)+H(Y)-H(X,Y)-\xi-\eta-\zeta)} \\
&\quad = 2^{-n(I(X;Y)-\xi-\eta-\zeta)} \\
&\quad = 2^{-n(I(X;Y)-\tau)}
\end{aligned}$$

An interpretation of Lemma 7.17



- Randomly choose a row with uniform distribution and randomly choose a column with uniform distribution.
-

$$\Pr\{\text{Obtaining a jointly typical pair}\} \approx \frac{2^{nH(X,Y)}}{2^{nH(X)}2^{nH(Y)}} = 2^{-nI((X;Y)}$$

Random Coding Scheme

- Fix $\epsilon > 0$ and input distribution $p(x)$. Let δ to be specified later.
- Let M be an even integer satisfying

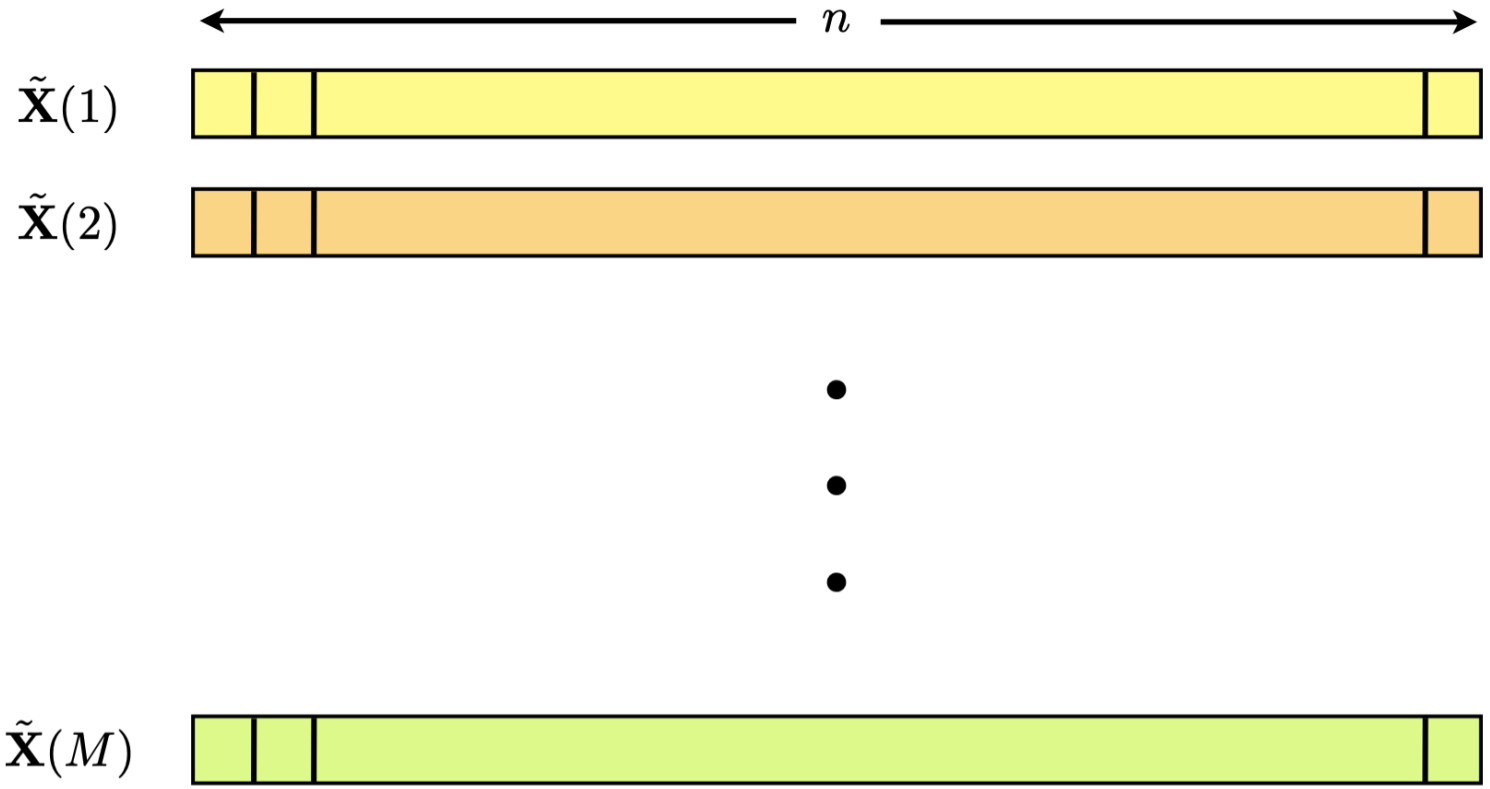
$$I(X;Y) - \frac{\epsilon}{2} < \frac{1}{n} \log M < I(X;Y) - \frac{\epsilon}{4},$$

  where n is sufficiently large, i.e., $M \approx 2^{nI(X;Y)}$.

The random coding scheme:

1. Construct the codebook $\mathcal{C}$ of an (n, M) code by generating M codewords in $\mathcal{X}^n$ independently and identically according to $p(x)^n$. Denote these codewords by $\tilde{\mathbf{X}}(1), \tilde{\mathbf{X}}(2), \cdots, \tilde{\mathbf{X}}(M)$.
2. Reveal the codebook $\mathcal{C}$ to both the encoder and the decoder.
3. A message W is chosen from $\mathcal{W}$ according to the uniform distribution.
4. Transmit $\mathbf{X} = \tilde{\mathbf{X}}(W)$ through the channel.

Random Coding Scheme



- Generate each component according to $p(x)$.
- There are a total of $|\mathcal{X}|^{Mn}$ possible codebooks that can be constructed.
- Regard two codebooks whose sets of codewords are permutations of each other as two different codebooks.

Random Coding Scheme

5. The channel outputs a sequence $\mathbf{Y}$ according to

$$\Pr\{\mathbf{Y} = \mathbf{y} | \tilde{\mathbf{X}}(W) = \mathbf{x}\} = \prod_{i=1}^{n} p(y_i|x_i)$$

6. The sequence $\mathbf{Y}$ is decoded to the message w if

   - $(\tilde{\mathbf{X}}(w), \mathbf{Y}) \in T^n_{[XY]\delta}$, and
   - there does not exists $w' \neq w$ such that $(\tilde{\mathbf{X}}(w'), \mathbf{Y}) \in T^n_{[XY]\delta}$.

   Otherwise, $\mathbf{Y}$ is decoded to a constant message in $\mathcal{W}$. Denote by $\hat{W}$ the message to which $\mathbf{Y}$ is decoded.

Idea of Analysis

- Let n be large.
- $\Pr\{\tilde{\mathbf{X}}(1) \text{ jointly typical with } \mathbf{Y}\} \to 1$.
- For $i \neq 1$, $\Pr\{\tilde{\mathbf{X}}(i) \text{ jointly typical with } \mathbf{Y}\} \approx 2^{-nI(X;Y)}$.
- If $|\mathcal{C}| = M$ grows at a rate $< I(X;Y)$, then

$$\Pr\{\tilde{\mathbf{X}}(i) \text{ jointly typical with } \mathbf{Y} \text{ for some } i \neq 1 \}$$

  can be made arbitrarily small.
- Then $\Pr\{\hat{W} \neq W\}$ can be made arbitrarily small.

A Discussion

- The channel coding theorem says that an indefinitely long message can be communicated reliably through the channel when the block length $n \to \infty$. This is much stronger than BER $\to 0$.
- The direct part of the channel coding theorem is an existence proof (as opposed to a constructive proof).
- A randomly constructed code has the following issues:
  - Encoding and decoding are computationally prohibitive.
  - High storage requirements for encoder and decoder.
- Nevertheless, the direct part implies that when n is large, if the codewords are chosen randomly, most likely the code is good (Markov lemma).
- It also gives much insight into what a good code would look like.
- In particular, the repetition code is not a good code because the numbers of '0' and '1's in the codewords are not roughly the same.

Thank you!